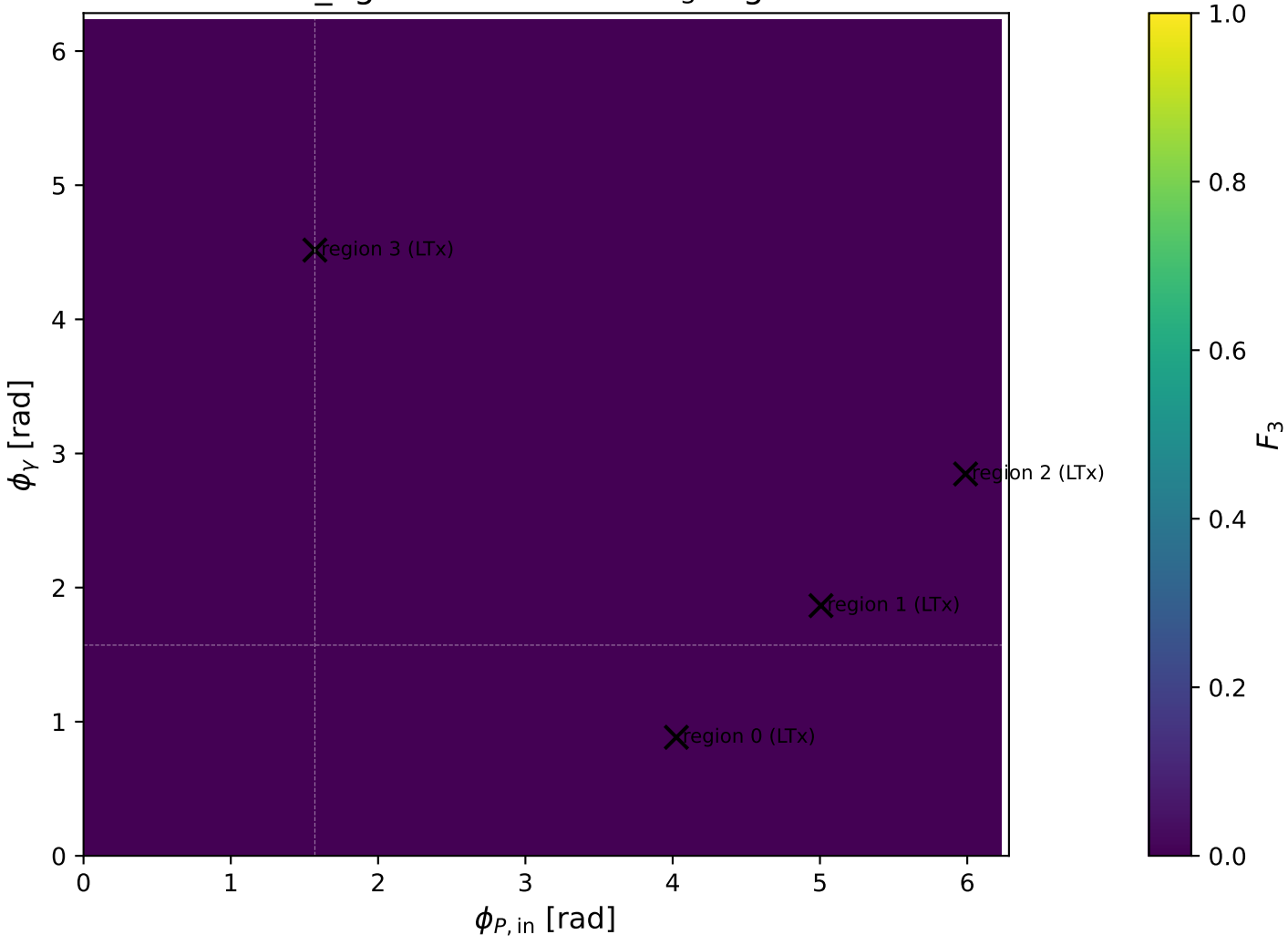
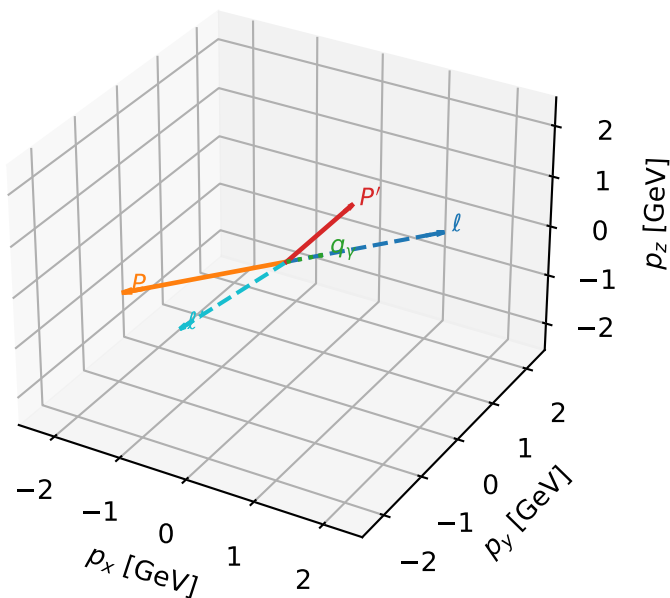


low\_Egamma LTx: max  $F_3$  regions



# LTx characteristic max $F_3$ configuration

3D momenta

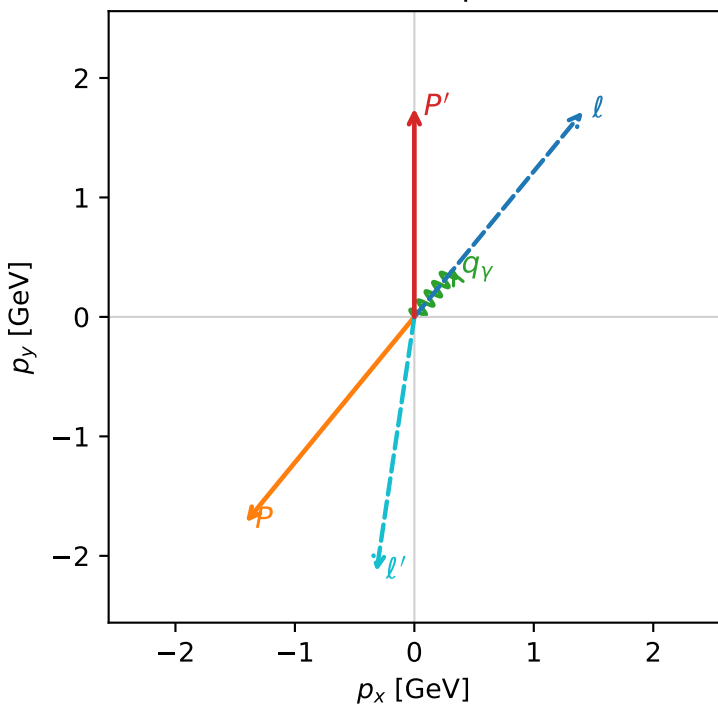


f3\_low\_egamma\_ltx\_region\_0 F\_3=0.00349817  
region: low\_Egamma\_LTx\_region\_0  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
incoming state:  $|L_e T_{x,p}\rangle$

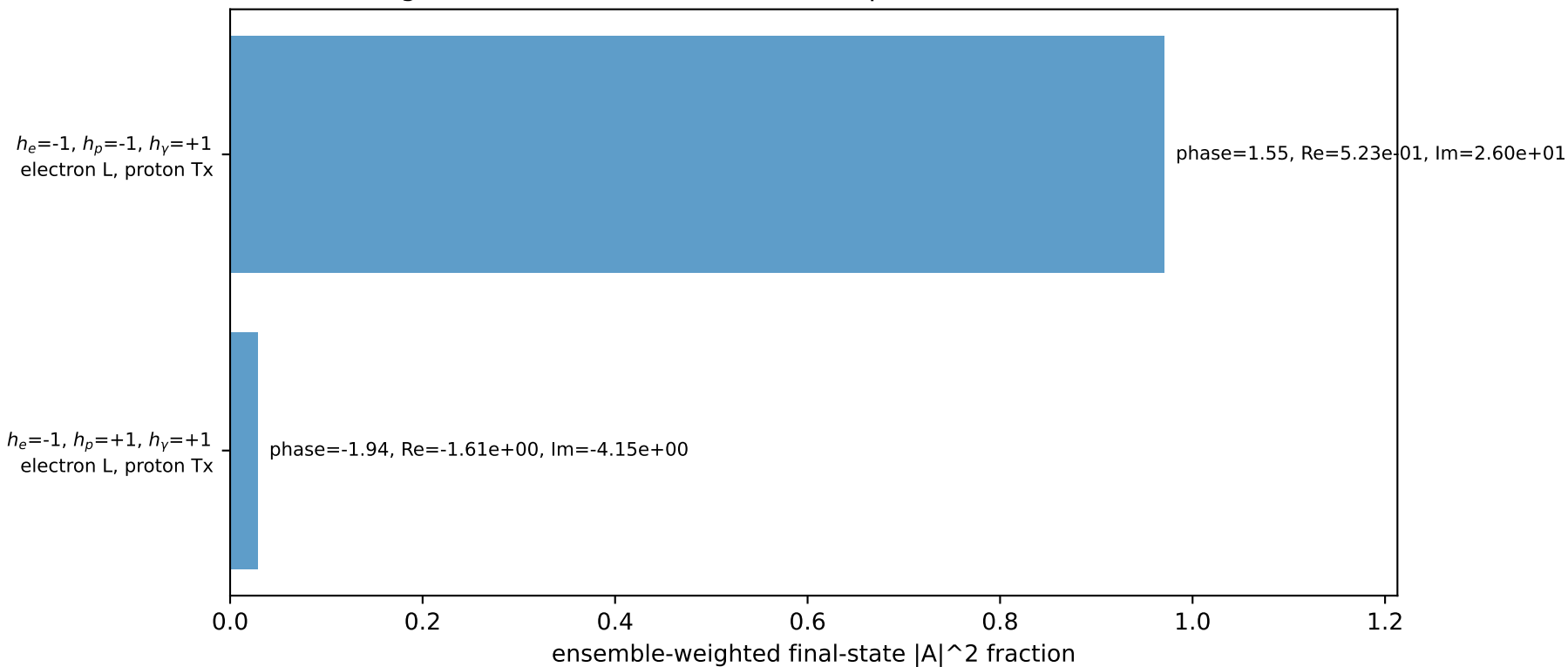
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.74629$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=0.883573$ ,  $\phi_P=4.02517$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=0.883573$   
 $Q^2=17.8227$ ,  $x_B=0.965735$ ,  $t=-13.8166$ ,  $W^2=1.51221$ ,  $y=0.894315$   
 $\theta(\ell', \gamma)=149.084$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 1.4112$   $p_y= 1.7195$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -1.4112$   $p_y= -1.7195$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1563$   $p_x= -0.3172$   $p_y= -2.1328$   $p_z= 0.0000$   
 $P'$   $E= 1.9823$   $p_x= 0.0000$   $p_y= 1.7463$   $p_z= 0.0000$   
 $q_Y$   $E= 0.5000$   $p_x= 0.3172$   $p_y= 0.3865$   $p_z= 0.0000$

Transverse plane

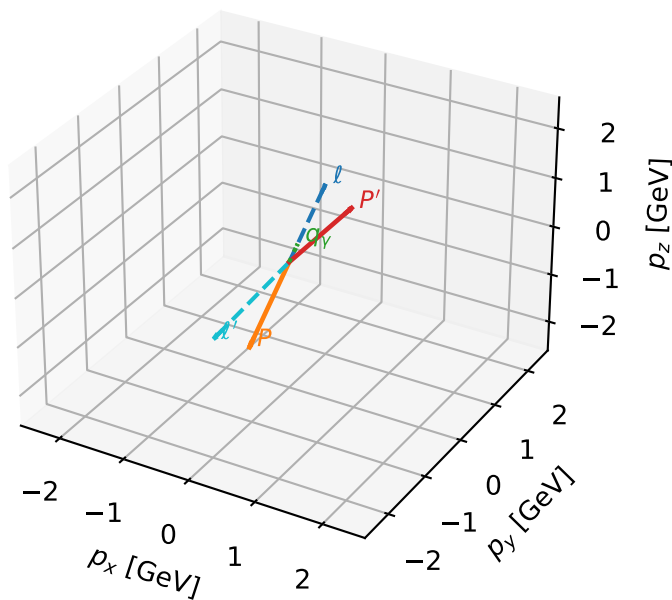


Leading initial-ensemble/final-state components (N=2, retained=99.9%)



# LTx characteristic max $F_3$ configuration

3D momenta

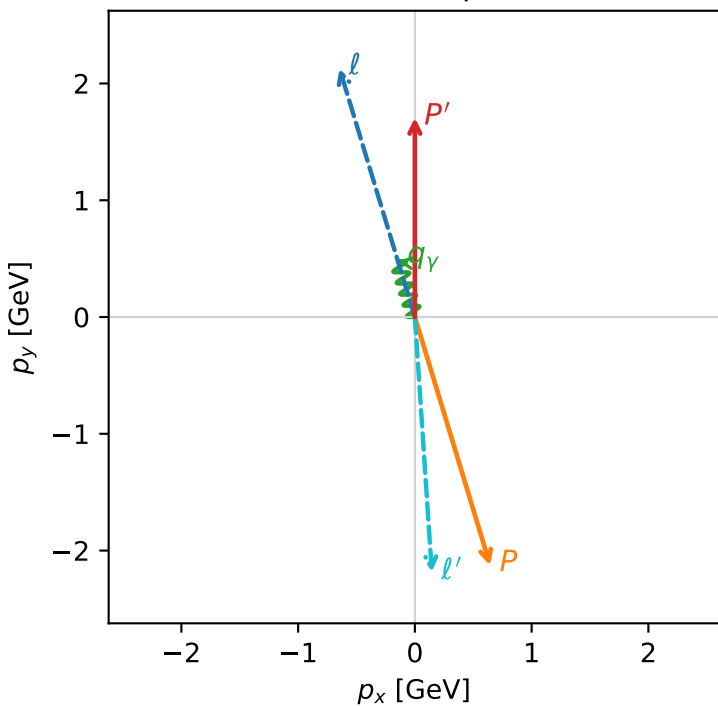


f3\_low\_egamma\_ltx\_region\_1 F\_3=0.00252281  
 region: low\_Egamma LTx\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_e T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.70726$   
 $\theta_{in}=1.57079$ ,  $\phi_l=1.86532$ ,  $\phi_p=5.00691$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=1.86532$   
 $Q^2=19.2381$ ,  $x_B=0.983929$ ,  $t=-14.9138$ ,  $W^2=1.19407$ ,  $y=0.947485$   
 $\theta(l', \gamma)=166.924$  deg

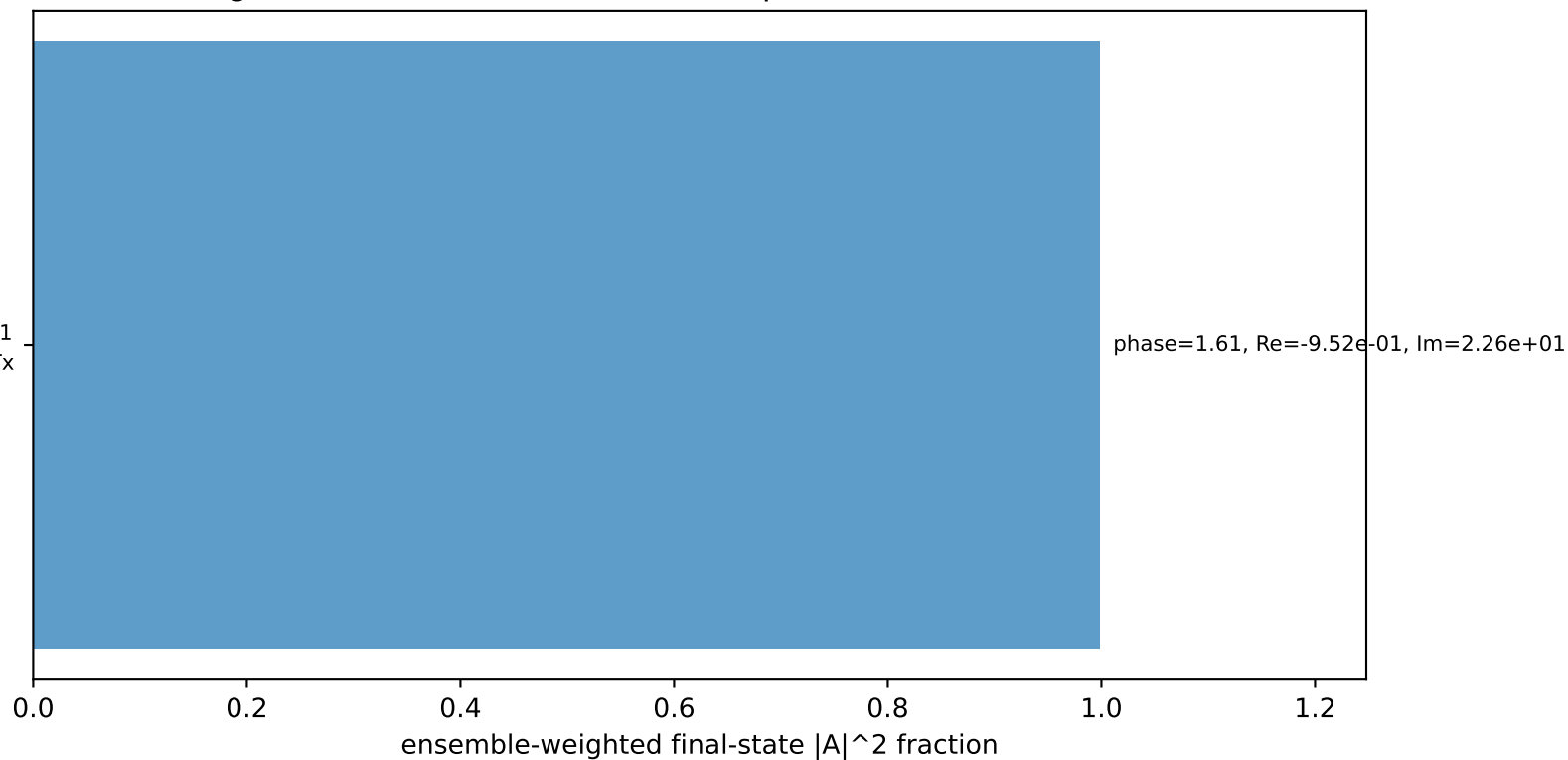
four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -0.6457$   $p_y= 2.1286$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.6457$   $p_y= -2.1286$   $p_z= 0.0000$   
 $l'$   $E= 2.1905$   $p_x= 0.1451$   $p_y= -2.1857$   $p_z= 0.0000$   
 $P'$   $E= 1.9480$   $p_x= 0.0000$   $p_y= 1.7073$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.5000$   $p_x= -0.1451$   $p_y= 0.4785$   $p_z= 0.0000$

Transverse plane



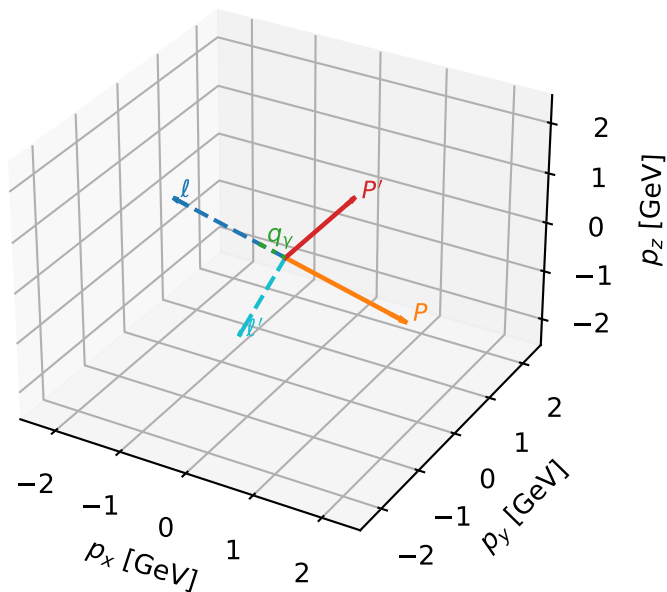
$h_e=-1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
 electron L, proton Tx

Leading initial-ensemble/final-state components (N=1, retained=99.8%)



# LTx characteristic max $F_3$ configuration

3D momenta

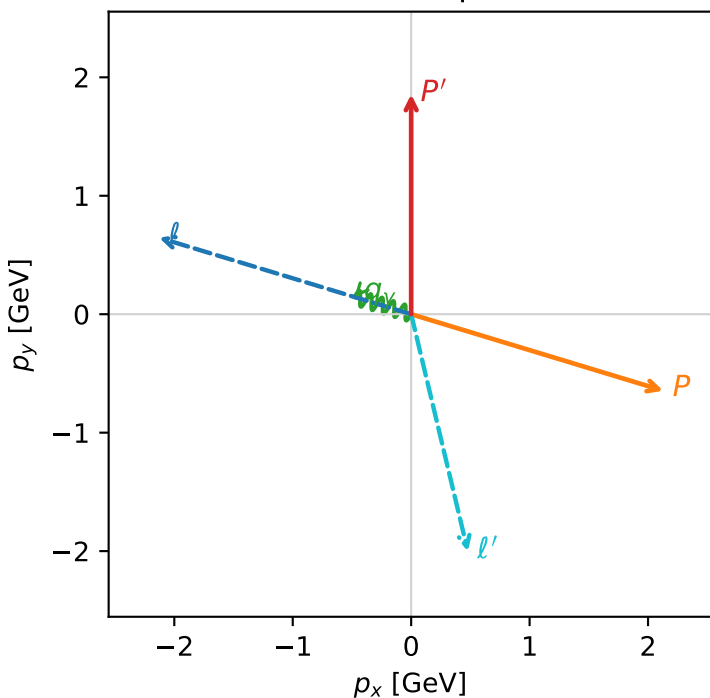


f3\_low\_egamma\_ltx\_region\_2 F\_3=0.00218253  
region: low\_Egamma\_LTx\_region\_2  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
incoming state:  $|L_e T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.85675$   
 $\theta_{in}=1.57079$ ,  $\phi_l=2.84707$ ,  $\phi_p=5.98866$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=2.84707$   
 $Q^2=13.7792$ ,  $x_B=0.8994$ ,  $t=-10.682$ ,  $W^2=2.42109$ ,  $y=0.742416$   
 $\theta(\ell', \gamma)=120.317$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -2.1286$   $p_y= 0.6457$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 2.1286$   $p_y= -0.6457$   $p_z= 0.0000$   
 $\ell'$   $E= 2.0583$   $p_x= 0.4785$   $p_y= -2.0019$   $p_z= 0.0000$   
 $P'$   $E= 2.0802$   $p_x= 0.0000$   $p_y= 1.8568$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.5000$   $p_x= -0.4785$   $p_y= 0.1451$   $p_z= 0.0000$

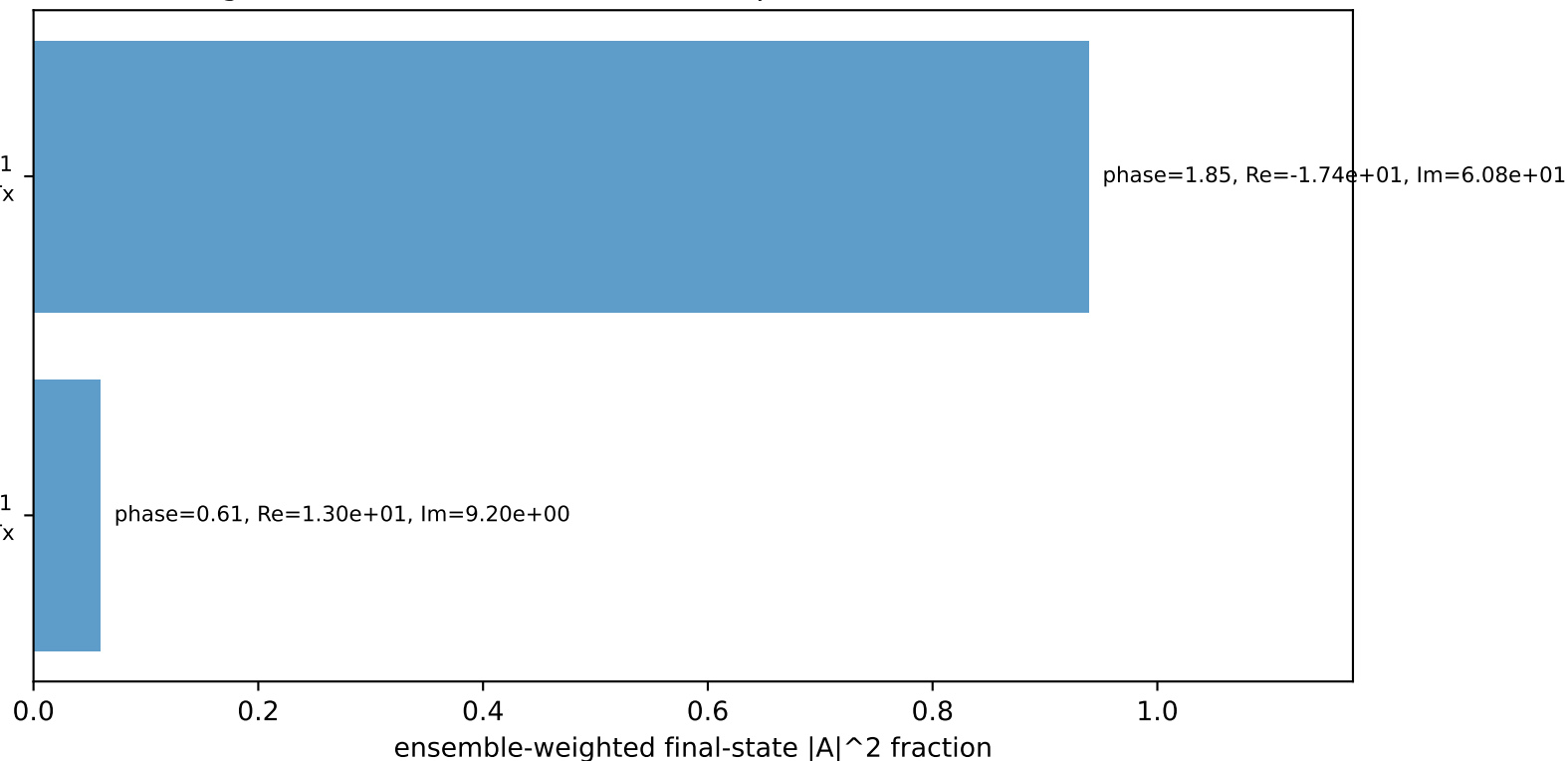
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$   
electron L, proton Tx

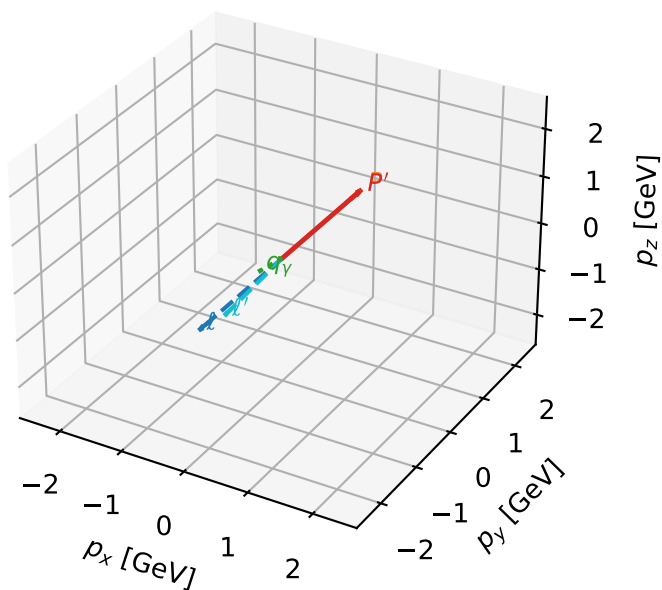
$h_e=-1, h_p=+1, h_\gamma=+1$   
electron L, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=99.9%)



# LTx characteristic max $F_3$ configuration

3D momenta

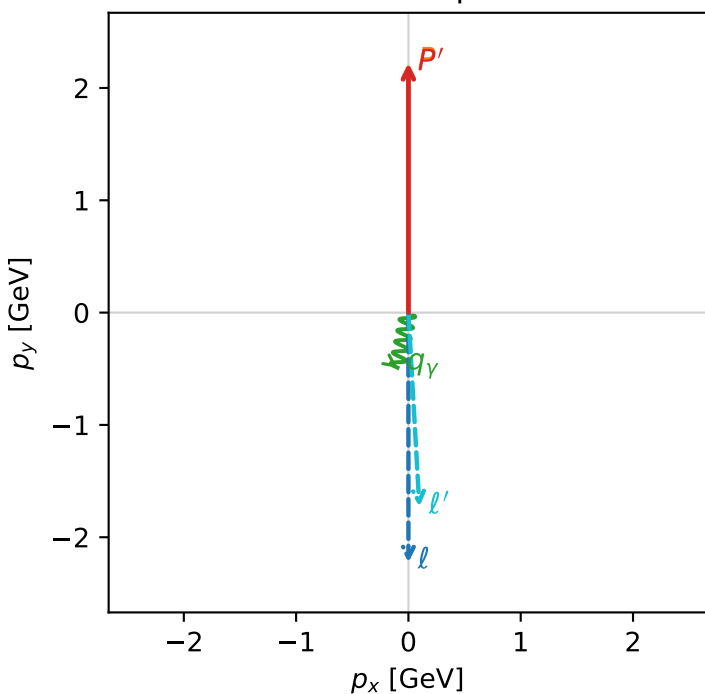


f3\_low\_egamma\_ltx\_region\_3 F\_3=0.00213094  
 region: low\_Egamma\_LTx\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_e T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.21798$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=4.51604$   
 $Q^2=0.0122417$ ,  $x_B=0.00266368$ ,  $t=-6.2629e-06$ ,  $W^2=5.46338$ ,  $y=0.222707$   
 $\theta(l', \gamma)=14.4817$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 1.7303$   $p_x= 0.0975$   $p_y= -1.7276$   $p_z= 0.0000$   
 $P'$   $E= 2.4082$   $p_x= 0.0000$   $p_y= 2.2180$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.5000$   $p_x= -0.0975$   $p_y= -0.4904$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=99.6%)

